MIT-ADT UNIVERSITY | MIT SCHOOL OF COMPUTING | DEPARTMENT OF CSE

ABSTRACT

An AI-Based Framework for Pre-Harvest Crop Yield and Production Estimation

Initial Case Study	Domain	Primary Task
Pomegranate (Punica granatum L.)	Agricultural Artificial Intelligence	Pre-harvest yield & production estimation

Agricultural decision-making often requires an estimate of crop production before the actual harvest, yet obtaining reliable pre-harvest yield information from standing fruit crops remains difficult. Manual fruit counting and field assessment are time-consuming, while image-based methods may observe only a portion of the crop because of foliage, occlusion, lighting variation, and limited spatial sampling. The literature reviewed for this study demonstrates several relevant capabilities: pomegranate growth-stage classification, lightweight fruit detection, repeated on-tree monitoring, and non-contact physical fruit-size estimation. However, within the selected seven-paper corpus, these capabilities are not integrated into a single validated framework that converts field-observed fruit information into tree-, row-, or orchard-level pre-harvest production estimates while explicitly considering incomplete visibility and cumulative estimation error. Therefore, this project proposes an AI-based framework for pre-harvest crop yield and total production estimation, with pomegranate as the initial case study. The proposed direction focuses on field image/video analysis, fruit detection and counting, extraction of relevant fruit-level features, crop-load estimation, production aggregation, and quantitative error or uncertainty analysis. The objective is to provide a practical and data-driven estimate of expected crop production before harvest, while keeping the core system software-oriented and suitable for M.Tech-level implementation.

Keywords

Pomegranate; Pre-harvest yield estimation; Crop production; Computer vision; Fruit detection; Crop-load estimation; Agricultural AI

Research Positioning

The literature-derived problem is framed as an integration gap within the selected seven-paper corpus, rather than a claim that crop-yield estimation has never been studied. The project focuses on connecting field-observed fruit information to quantitative pre-harvest production estimates.